

Mercury levels and fish consumption practices in women of child-bearing age in the Florida Panhandle.



The southeastern United States, and in particular the coastal areas along the Gulf of Mexico (Gulf Coast) in Florida, experience some of the highest levels of mercury deposition in the country. Although the State of Florida's coastal border is among the longest in the United States, and the State has issued fish consumption advisories due to mercury on multiple fish species, few data have been systematically collected to assess mercury levels in the human population of the state or to assess the efficacy of the consumption advisories. Because of the generally high rate of seafood consumption among coastal populations, the human population in the Florida Panhandle, near Pensacola, FL is potentially exposed to elevated levels of mercury. In the present study, we analyzed hair mercury levels in women of child-bearing age (16-49 years) who had resided near Pensacola, FL for at least 1 year. We also surveyed the fish consumption practices of the cohort and evaluated awareness of the Florida Fish Consumption Advisory. Hair mercury levels were significantly higher in women who consumed fish within the 30 days prior to sampling ($p < 0.05$) and in those women who were unaware of the consumption advisory ($p < 0.05$). Only 31% of the women reported knowledge of the consumption advisory and pregnant women exhibited lower awareness of the advisory than non-pregnant women. The data suggest that public health interventions such as education and fish advisories have not reached the majority of women in the counties surrounding Pensacola who are most at risk from consumption of fish with high levels of mercury.